Haq: help getting your money back

Problem 2, Build an AI Lawyer. Prototype: [LINK] · Code, test cases and raw outputs: https://github.com/mohitmethi1000/haq

What I picked and why

I picked this problem because its main risk, made-up law, can actually be tested. A citation is either real or it isn't.

Haq is for someone in India who is owed money and won't pay a lawyer to get it back. They type or speak in English, Hindi or Hinglish. It handles unpaid salary and consumer refunds, because both are central law and follow the same path: facts, deadline, where to file, demand letter. Rental deposits are only partly covered, since they depend on State law I didn't verify. Cheque bounce, family, property and anything else goes to free legal aid (15100) instead of a guess. Scams, violence, self-harm and minors are caught first and sent to the right helpline.

The government's Nyaya Setu bot already does voice in Indian languages, so voice alone isn't new. What I wanted was a correct deadline and a usable letter, with the error rate shown.

Assumptions. Central law only. No lawyer was available, so the statute text is my answer key: I can say the output matches the text, not that it is good legal advice. The user checks the facts and sends any letter. I call it a legal information tool, not a lawyer.

How it works

Indian law changed recently. The Code on Wages started on 21 Nov 2025 and BNS replaced the IPC on 1 Jul 2024. So besides inventing law, a model is likely to cite real law that no longer applies.

Step	Done by
Turn the story into facts (unknown stays blank)	AI
Check and correct those facts	**Person**
Spot scams, violence, self-harm, minors	Code
Deadlines, where to file, fees, whether the wage law covers this worker	Code
Explain in plain language, using only a small set of verified sections	AI
Check every citation; retry once, then remove what fails and send to a person	Code
Check whether each cited section supports its sentence	AI, as a warning only
Translate; reject if any number changes	AI + code
Fill the letter from confirmed facts; blanks are marked	Code
Edit, sign and send	**Person**

How it avoids invented law. It can only cite 32 provisions I took from official text (26 from the Gazette, 3 from Indian Kanoon, 3 summaries of rules), each with its source and date. The output format has no field for court cases. Code then rejects any section or Act not on the list, anything that looks like a case, repealed law unless it is being corrected, and any time limit, percentage or date that isn't in the law, the computed rules or the user's facts. If a sentence can't be confirmed, it is shown separately and the user is told to check it.

There is no vector database or agent framework, because 32 provisions fit in the prompt and the steps are fixed. Changing the model takes a base URL, a key and a model name. The default is Sarvam-105B, and the checks stay in code whichever model is used.

How I know it's good, and where it fails

I wrote 50 test cases: normal cases, old-law traps, fake sections, out-of-scope questions, safety, prompt injection, one fact changed at a time, and the same story in four language forms. A separate set of 22 cases (26 runs) was written by another AI agent and not used for tuning before its recorded run. All of it is graded by code, and the unedited outputs are in `raw_outputs/`.

Same model and same questions, four ways. "Careful" is a prompt that gives the date and the law changes and says to cite only what it's sure of.

Answers containing... (main set, 26)	Plain chatbot	Careful	Haq before checks	Haq
Section numbers with no verified source	25	11	4	0
Court cases or rulings	16	2	1	0
Repealed law, not corrected	18	9	3	0
Second set (16): same three rows	*16, 7, 10*	*4, 0, 8*	*3, 0, 3*	*1, 0, 2*

The Haq column uses patterns close to the ones the checks enforce, so the "before checks" column is the fairer evidence.

Overall, 44 of 50 main cases passed, and 3 of the 6 failures were mistakes in my grader. **The second set passed 19 of 26**, and that is the number I trust. The gap shows I had partly tuned to my own cases. Some fixes came after these runs and aren't re-measured.

Where it fails:

- A real section cited for the wrong claim passes the code checks. The AI check catches some, but it flagged 33% of citations on real answers and many of those were false alarms.

- Facts go wrong in non-English input, like a date in the wrong field. Only the person checking can catch that.
- The checks sometimes remove a correct correction.
- It didn't always deny a fake section.
- No lawyer or native speaker has reviewed it. Voice was only tested on a synthetic clip. Switching models hasn't been tested with a second paid provider.

Scale and cost

Each case took 3.3 model calls, about ₹0.22 and 10 seconds on Sarvam. At an assumed 1 million conversations a month, that is roughly ₹2.2 lakh for one-pass text, about ₹12 lakh with voice input and about ₹32 lakh with a short spoken answer, at today's list prices, before hosting and staff.

Tokens aren't the limit, people are. 19 of 45 in-scope main cases were sent to a person, many by design. Even 10% of a million is about 100,000 handoffs a month, and Tele-Law handles roughly 33,000. So the next work is verified State deposit rules and better self-serve letters. A real launch also needs a lawyer reviewing the rules, a process for law changes, permission to use India Code text, and data protection compliance.

AI tools, and one place they got it wrong

Claude Code did most of the research and coding, working from my direction. Separate Claude agents reviewed the work and wrote the second test set. Sarvam-105B is the product's model and Sarvam Saaras does speech to text.

The citation check Claude wrote only knew English correction words. In the first real run, the model correctly said "Payment of Wages Act khatam ho gaya, iski jagah Code on Wages aa gaya", and the check deleted that sentence as citing repealed law. Reading the raw output caught it. The first fix went too far and let words like "know" count as corrections, so now a sentence must name the replacement law. A review also found a worse error: the rules applied the Code on Wages to wages that fell due before it existed. So I stopped trusting a passing test on its own.